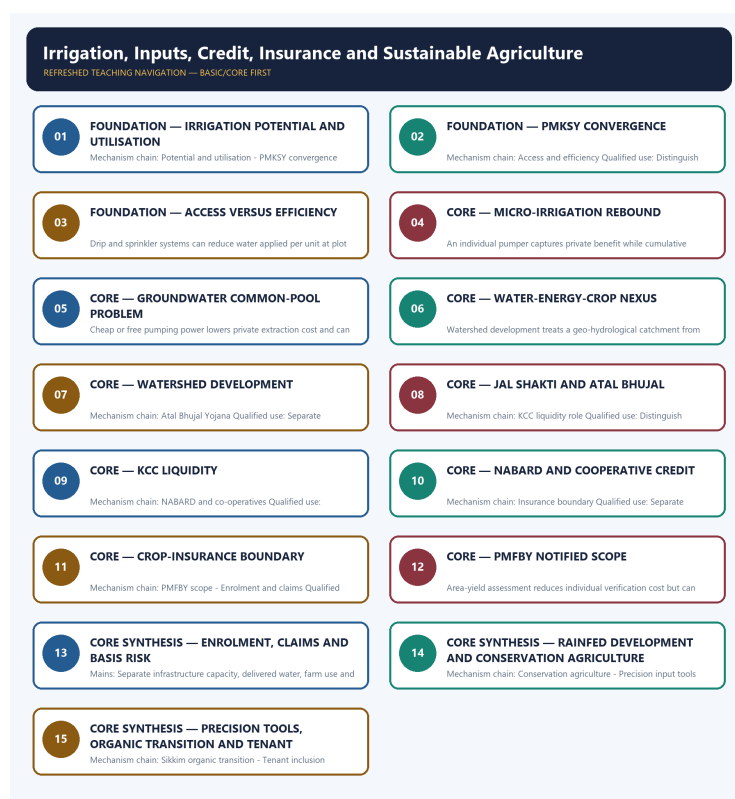


## SOLVED PRACTICE WORKBOOK

# Irrigation, Inputs, Credit, Insurance and Sustainable Agriculture - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

# CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

## Irrigation, Inputs, Credit, Insurance and Sustainable Agriculture - Solved Practice Workbook . . . . 6

### BASIC MCQS / REMEDIATION . . . . . 6

|                                                                                                                  |    |
|------------------------------------------------------------------------------------------------------------------|----|
| Q1. Which statement correctly identifies Potential and utilisation? . . . . .                                    | 6  |
| Q2. Which option preserves the accounting or regulatory boundary of Potential and utilisation? . . . . .         | 6  |
| Q3. Which statement uses Potential and utilisation without losing its vintage, basket or legal status? . . . . . | 6  |
| Q4. Which option avoids the standard UPSC close-option trap about Potential and utilisation? . . . . .           | 6  |
| Q5. Which statement correctly identifies PMKSY convergence? . . . . .                                            | 7  |
| Q6. Which option preserves the accounting or regulatory boundary of PMKSY convergence? . . . . .                 | 7  |
| Q7. Which statement uses PMKSY convergence without losing its vintage, basket or legal status? . . . . .         | 7  |
| Q8. Which option avoids the standard UPSC close-option trap about PMKSY convergence? . . . . .                   | 7  |
| Q9. Which statement correctly identifies Access and efficiency? . . . . .                                        | 8  |
| Q10. Which option preserves the accounting or regulatory boundary of Access and efficiency? . . . . .            | 8  |
| Q11. Which statement uses Access and efficiency without losing its vintage, basket or legal status? . . . . .    | 8  |
| Q12. Which option avoids the standard UPSC close-option trap about Access and efficiency? . . . . .              | 8  |
| Q13. Which statement correctly identifies Micro-irrigation rebound? . . . . .                                    | 9  |
| Q14. Which option preserves the accounting or regulatory boundary of Micro-irrigation rebound? . . . . .         | 9  |
| Q15. Which statement uses Micro-irrigation rebound without losing its vintage, basket or legal status? . . . . . | 9  |
| Q16. Which option avoids the standard UPSC close-option trap about Micro-irrigation rebound? . . . . .           | 9  |
| Q17. Which statement correctly identifies Groundwater common pool? . . . . .                                     | 10 |
| Q18. Which option preserves the accounting or regulatory boundary of Groundwater common pool? . . . . .          | 10 |
| Q19. Which statement uses Groundwater common pool without losing its vintage, basket or legal status? . . . . .  | 10 |
| Q20. Which option avoids the standard UPSC close-option trap about Groundwater common pool? . . . . .            | 10 |
| Q21. Which statement correctly identifies Water-energy-crop nexus? . . . . .                                     | 11 |
| Q22. Which option preserves the accounting or regulatory boundary of Water-energy-crop nexus? . . . . .          | 11 |
| Q23. Which statement uses Water-energy-crop nexus without losing its vintage, basket or legal status? . . . . .  | 11 |
| Q24. Which option avoids the standard UPSC close-option trap about Water-energy-crop nexus? . . . . .            | 11 |
| Q25. Which statement correctly identifies Watershed method? . . . . .                                            | 12 |
| Q26. Which option preserves the accounting or regulatory boundary of Watershed method? . . . . .                 | 12 |
| Q27. Which statement uses Watershed method without losing its vintage, basket or legal status? . . . . .         | 12 |

|                                                                                                                  |    |
|------------------------------------------------------------------------------------------------------------------|----|
| Q28. Which option avoids the standard UPSC close-option trap about Watershed method? . . . . .                   | 12 |
| Q29. Which statement correctly identifies Jal Shakti campaign? . . . . .                                         | 13 |
| Q30. Which option preserves the accounting or regulatory boundary of Jal Shakti campaign? . . . . .              | 13 |
| Q31. Which statement uses Jal Shakti campaign without losing its vintage, basket or legal status? . . . . .      | 13 |
| Q32. Which option avoids the standard UPSC close-option trap about Jal Shakti campaign? . . . . .                | 13 |
| Q33. Which statement correctly identifies Atal Bhujal Yojana? . . . . .                                          | 14 |
| Q34. Which option preserves the accounting or regulatory boundary of Atal Bhujal Yojana? . . . . .               | 14 |
| Q35. Which statement uses Atal Bhujal Yojana without losing its vintage, basket or legal status? . . . . .       | 14 |
| Q36. Which option avoids the standard UPSC close-option trap about Atal Bhujal Yojana? . . . . .                 | 14 |
| Q37. Which statement correctly identifies KCC liquidity role? . . . . .                                          | 15 |
| Q38. Which option preserves the accounting or regulatory boundary of KCC liquidity role? . . . . .               | 15 |
| Q39. Which statement uses KCC liquidity role without losing its vintage, basket or legal status? . . . . .       | 15 |
| Q40. Which option avoids the standard UPSC close-option trap about KCC liquidity role? . . . . .                 | 15 |
| Q41. Which statement correctly identifies NABARD and co-operatives? . . . . .                                    | 16 |
| Q42. Which option preserves the accounting or regulatory boundary of NABARD and co-operatives? . . . . .         | 16 |
| Q43. Which statement uses NABARD and co-operatives without losing its vintage, basket or legal status? . . . . . | 16 |
| Q44. Which option avoids the standard UPSC close-option trap about NABARD and co-operatives? . . . . .           | 16 |
| Q45. Which statement correctly identifies Insurance boundary? . . . . .                                          | 17 |
| Q46. Which option preserves the accounting or regulatory boundary of Insurance boundary? . . . . .               | 17 |
| Q47. Which statement uses Insurance boundary without losing its vintage, basket or legal status? . . . . .       | 17 |
| Q48. Which option avoids the standard UPSC close-option trap about Insurance boundary? . . . . .                 | 17 |
| Q49. Which statement correctly identifies PMFBY scope? . . . . .                                                 | 18 |
| Q50. Which option preserves the accounting or regulatory boundary of PMFBY scope? . . . . .                      | 18 |
| Q51. Which statement uses PMFBY scope without losing its vintage, basket or legal status? . . . . .              | 18 |
| Q52. Which option avoids the standard UPSC close-option trap about PMFBY scope? . . . . .                        | 18 |
| Q53. Which statement correctly identifies Enrolment and claims? . . . . .                                        | 19 |
| Q54. Which option preserves the accounting or regulatory boundary of Enrolment and claims? . . . . .             | 19 |
| Q55. Which statement uses Enrolment and claims without losing its vintage, basket or legal status? . . . . .     | 19 |
| Q56. Which option avoids the standard UPSC close-option trap about Enrolment and claims? . . . . .               | 19 |
| Q57. Which statement correctly identifies Basis risk? . . . . .                                                  | 20 |
| Q58. Which option preserves the accounting or regulatory boundary of Basis risk? . . . . .                       | 20 |
| Q59. Which statement uses Basis risk without losing its vintage, basket or legal status? . . . . .               | 20 |

|                                                                                                                   |           |
|-------------------------------------------------------------------------------------------------------------------|-----------|
| Q60. Which option avoids the standard UPSC close-option trap about Basis risk? . . . . .                          | 20        |
| Q61. Which statement correctly identifies Rainfed Area Development? . . . . .                                     | 21        |
| Q62. Which option preserves the accounting or regulatory boundary of Rainfed Area Development? . . . . .          | 21        |
| Q63. Which statement uses Rainfed Area Development without losing its vintage, basket or legal status? . . . . .  | 21        |
| Q64. Which option avoids the standard UPSC close-option trap about Rainfed Area Development? . . . . .            | 21        |
| Q65. Which statement correctly identifies Conservation agriculture? . . . . .                                     | 22        |
| Q66. Which option preserves the accounting or regulatory boundary of Conservation agriculture? . . . . .          | 22        |
| Q67. Which statement uses Conservation agriculture without losing its vintage, basket or legal status? . . . . .  | 22        |
| Q68. Which option avoids the standard UPSC close-option trap about Conservation agriculture? . . . . .            | 22        |
| Q69. Which statement correctly identifies Precision input tools? . . . . .                                        | 23        |
| Q70. Which option preserves the accounting or regulatory boundary of Precision input tools? . . . . .             | 23        |
| Q71. Which statement uses Precision input tools without losing its vintage, basket or legal status? . . . . .     | 23        |
| Q72. Which option avoids the standard UPSC close-option trap about Precision input tools? . . . . .               | 23        |
| Q73. Which statement correctly identifies Sikkim organic transition? . . . . .                                    | 24        |
| Q74. Which option preserves the accounting or regulatory boundary of Sikkim organic transition? . . . . .         | 24        |
| Q75. Which statement uses Sikkim organic transition without losing its vintage, basket or legal status? . . . . . | 24        |
| Q76. Which option avoids the standard UPSC close-option trap about Sikkim organic transition? . . . . .           | 24        |
| Q77. Which statement correctly identifies Tenant inclusion? . . . . .                                             | 25        |
| Q78. Which option preserves the accounting or regulatory boundary of Tenant inclusion? . . . . .                  | 25        |
| Q79. Which statement uses Tenant inclusion without losing its vintage, basket or legal status? . . . . .          | 25        |
| Q80. Which option avoids the standard UPSC close-option trap about Tenant inclusion? . . . . .                    | 25        |
| <b>PYQS AND ANSWER PRACTICE . . . . .</b>                                                                         | <b>26</b> |
| VERIFIED PYQ OWNERSHIP AUDIT . . . . .                                                                            | 26        |
| OWNER PYQ LEDGER EXTRACTS . . . . .                                                                               | 26        |
| PYQ DEMAND CARD 1 - 2018 GS-III . . . . .                                                                         | 32        |
| PYQ DEMAND CARD 2 - 2019 GS-III . . . . .                                                                         | 33        |
| PYQ DEMAND CARD 3 - 2020 GS-III . . . . .                                                                         | 34        |
| PYQ DEMAND CARD 4 - 2021 GS-III . . . . .                                                                         | 35        |
| PYQ DEMAND CARD 5 - 2024 GS-III . . . . .                                                                         | 36        |
| PYQ DEMAND CARD 6 - 2025 GS-III . . . . .                                                                         | 38        |
| ORIGINAL MAINS 1 - 10 MARKS . . . . .                                                                             | 39        |
| ORIGINAL MAINS 2 - 10 MARKS . . . . .                                                                             | 40        |

ORIGINAL MAINS 3 - 15 MARKS . . . . . 42

ORIGINAL MAINS 4 - 15 MARKS . . . . . 44

ORIGINAL MAINS 5 - 20 MARKS . . . . . 47

ORIGINAL MAINS 6 - 20 MARKS . . . . . 51

# Irrigation, Inputs, Credit, Insurance and Sustainable Agriculture - Solved Practice Workbook

## BASIC MCQS / REMEDIATION

### Q1. Which statement correctly identifies Potential and utilisation?

A. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. B. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. C. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. D. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning.

**Answer: A. Explanation: Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q2. Which option preserves the accounting or regulatory boundary of Potential and utilisation?

A. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. B. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. C. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. D. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged.

**Answer: B. Explanation: Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q3. Which statement uses Potential and utilisation without losing its vintage, basket or legal status?

A. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. B. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. C. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. D. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive.

**Answer: C. Explanation: Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q4. Which option avoids the standard UPSC close-option trap about Potential and utilisation?

A. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. B. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. C. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. D. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use.

**Answer: D. Explanation: Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. The other**

options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.

**Q5. Which statement correctly identifies PMKSY convergence?**

A. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. B. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. C. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. D. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem.

**Answer: A. Explanation: The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

**Q6. Which option preserves the accounting or regulatory boundary of PMKSY convergence?**

A. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. B. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. C. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. D. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive.

**Answer: B. Explanation: The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

**Q7. Which statement uses PMKSY convergence without losing its vintage, basket or legal status?**

A. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. B. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. C. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. D. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction.

**Answer: C. Explanation: The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

**Q8. Which option avoids the standard UPSC close-option trap about PMKSY convergence?**

A. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. B. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. C. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. D. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning.

**Answer: D. Explanation: The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q9. Which statement correctly identifies Access and efficiency?

A. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. B. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. C. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. D. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction.

**Answer: A. Explanation: Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q10. Which option preserves the accounting or regulatory boundary of Access and efficiency?

A. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. B. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. C. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. D. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction.

**Answer: B. Explanation: Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q11. Which statement uses Access and efficiency without losing its vintage, basket or legal status?

A. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. B. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. C. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. D. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction.

**Answer: C. Explanation: Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q12. Which option avoids the standard UPSC close-option trap about Access and efficiency?

A. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. B. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. C. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. D. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged.

**Answer: D. Explanation: Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q13. Which statement correctly identifies Micro-irrigation rebound?

A. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. B. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. C. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. D. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction.

**Answer: A. Explanation: Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q14. Which option preserves the accounting or regulatory boundary of Micro-irrigation rebound?

A. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. B. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. C. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. D. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures.

**Answer: B. Explanation: Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q15. Which statement uses Micro-irrigation rebound without losing its vintage, basket or legal status?

A. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. B. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. C. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. D. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change.

**Answer: C. Explanation: Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q16. Which option avoids the standard UPSC close-option trap about Micro-irrigation rebound?

A. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. B. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. C. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. D. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive.

**Answer: D. Explanation: Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q17. Which statement correctly identifies Groundwater common pool?

A. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. B. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. C. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. D. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures.

**Answer: A. Explanation: An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q18. Which option preserves the accounting or regulatory boundary of Groundwater common pool?

A. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. B. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. C. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. D. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record.

**Answer: B. Explanation: An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q19. Which statement uses Groundwater common pool without losing its vintage, basket or legal status?

A. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. B. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. C. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. D. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss.

**Answer: C. Explanation: An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q20. Which option avoids the standard UPSC close-option trap about Groundwater common pool?

A. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. B. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. C. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. D. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem.

**Answer: D. Explanation: An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q21. Which statement correctly identifies Water-energy-crop nexus?

A. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. B. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. C. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. D. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures.

**Answer: A. Explanation: Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q22. Which option preserves the accounting or regulatory boundary of Water-energy-crop nexus?

A. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. B. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. C. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. D. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss.

**Answer: B. Explanation: Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q23. Which statement uses Water-energy-crop nexus without losing its vintage, basket or legal status?

A. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. B. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. C. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. D. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change.

**Answer: C. Explanation: Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q24. Which option avoids the standard UPSC close-option trap about Water-energy-crop nexus?

A. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. B. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. C. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. D. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction.

**Answer: D. Explanation: Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q25. Which statement correctly identifies Watershed method?

A. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. B. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. C. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. D. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss.

**Answer: A. Explanation: Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q26. Which option preserves the accounting or regulatory boundary of Watershed method?

A. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. B. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. C. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. D. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture.

**Answer: B. Explanation: Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q27. Which statement uses Watershed method without losing its vintage, basket or legal status?

A. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. B. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. C. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. D. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss.

**Answer: C. Explanation: Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q28. Which option avoids the standard UPSC close-option trap about Watershed method?

A. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. B. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. C. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. D. Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures.

**Answer: D. Explanation: Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q29. Which statement correctly identifies Jal Shakti campaign?

A. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. B. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. C. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. D. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture.

**Answer: A. Explanation: The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q30. Which option preserves the accounting or regulatory boundary of Jal Shakti campaign?

A. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. B. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. C. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. D. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss.

**Answer: B. Explanation: The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q31. Which statement uses Jal Shakti campaign without losing its vintage, basket or legal status?

A. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. B. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. C. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. D. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification.

**Answer: C. Explanation: The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q32. Which option avoids the standard UPSC close-option trap about Jal Shakti campaign?

A. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. B. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. C. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. D. The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record.

**Answer: D. Explanation: The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q33. Which statement correctly identifies Atal Bhujal Yojana?

A. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. B. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. C. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. D. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss.

**Answer: A. Explanation: Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q34. Which option preserves the accounting or regulatory boundary of Atal Bhujal Yojana?

A. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. B. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. C. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. D. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss.

**Answer: B. Explanation: Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q35. Which statement uses Atal Bhujal Yojana without losing its vintage, basket or legal status?

A. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. B. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. C. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. D. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss.

**Answer: C. Explanation: Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q36. Which option avoids the standard UPSC close-option trap about Atal Bhujal Yojana?

A. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. B. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. C. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. D. Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change.

**Answer: D. Explanation: Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q37. Which statement correctly identifies KCC liquidity role?

A. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. B. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. C. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. D. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification.

**Answer: A. Explanation: Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q38. Which option preserves the accounting or regulatory boundary of KCC liquidity role?

A. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. B. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. C. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. D. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification.

**Answer: B. Explanation: Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q39. Which statement uses KCC liquidity role without losing its vintage, basket or legal status?

A. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. B. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. C. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. D. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification.

**Answer: C. Explanation: Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q40. Which option avoids the standard UPSC close-option trap about KCC liquidity role?

A. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. B. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. C. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. D. Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss.

**Answer: D. Explanation: Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q41. Which statement correctly identifies NABARD and co-operatives?

A. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. B. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. C. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. D. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss.

**Answer: A. Explanation: NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q42. Which option preserves the accounting or regulatory boundary of NABARD and co-operatives?

A. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. B. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. C. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. D. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes.

**Answer: B. Explanation: NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q43. Which statement uses NABARD and co-operatives without losing its vintage, basket or legal status?

A. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. B. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. C. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. D. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes.

**Answer: C. Explanation: NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q44. Which option avoids the standard UPSC close-option trap about NABARD and co-operatives?

A. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. B. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. C. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. D. NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture.

**Answer: D. Explanation: NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q45. Which statement correctly identifies Insurance boundary?

A. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. B. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. C. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. D. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes.

**Answer: A. Explanation: Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q46. Which option preserves the accounting or regulatory boundary of Insurance boundary?

A. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. B. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. C. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. D. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk.

**Answer: B. Explanation: Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q47. Which statement uses Insurance boundary without losing its vintage, basket or legal status?

A. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. B. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. C. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. D. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss.

**Answer: C. Explanation: Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q48. Which option avoids the standard UPSC close-option trap about Insurance boundary?

A. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. B. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. C. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. D. Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss.

**Answer: D. Explanation: Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q49. Which statement correctly identifies PMFBY scope?

A. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. B. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. C. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. D. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes.

**Answer: A. Explanation: PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q50. Which option preserves the accounting or regulatory boundary of PMFBY scope?

A. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. B. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. C. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. D. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss.

**Answer: B. Explanation: PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q51. Which statement uses PMFBY scope without losing its vintage, basket or legal status?

A. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. B. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. C. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. D. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system.

**Answer: C. Explanation: PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

#### Q52. Which option avoids the standard UPSC close-option trap about PMFBY scope?

A. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. B. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. C. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. D. PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification.

**Answer: D. Explanation: PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

**Q53. Which statement correctly identifies Enrolment and claims?**

A. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. B. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. C. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. D. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss.

**Answer: A. Explanation: Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

**Q54. Which option preserves the accounting or regulatory boundary of Enrolment and claims?**

A. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. B. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. C. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. D. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system.

**Answer: B. Explanation: Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

**Q55. Which statement uses Enrolment and claims without losing its vintage, basket or legal status?**

A. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. B. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. C. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. D. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system.

**Answer: C. Explanation: Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

**Q56. Which option avoids the standard UPSC close-option trap about Enrolment and claims?**

A. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. B. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. C. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. D. Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes.

**Answer: D. Explanation: Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q57. Which statement correctly identifies Basis risk?

A. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. B. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. C. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. D. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument.

**Answer: A. Explanation: Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q58. Which option preserves the accounting or regulatory boundary of Basis risk?

A. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. B. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. C. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. D. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument.

**Answer: B. Explanation: Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q59. Which statement uses Basis risk without losing its vintage, basket or legal status?

A. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. B. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. C. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. D. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument.

**Answer: C. Explanation: Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q60. Which option avoids the standard UPSC close-option trap about Basis risk?

A. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. B. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. C. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. D. Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss.

**Answer: D. Explanation: Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q61. Which statement correctly identifies Rainfed Area Development?

A. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. B. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. C. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. D. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication.

**Answer: A. Explanation: Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q62. Which option preserves the accounting or regulatory boundary of Rainfed Area Development?

A. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. B. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. C. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. D. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter.

**Answer: B. Explanation: Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q63. Which statement uses Rainfed Area Development without losing its vintage, basket or legal status?

A. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. B. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. C. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. D. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication.

**Answer: C. Explanation: Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q64. Which option avoids the standard UPSC close-option trap about Rainfed Area Development?

A. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. B. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. C. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. D. Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk.

**Answer: D. Explanation: Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q65. Which statement correctly identifies Conservation agriculture?

A. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. B. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. C. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. D. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter.

**Answer: A. Explanation: Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q66. Which option preserves the accounting or regulatory boundary of Conservation agriculture?

A. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. B. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. C. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. D. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use.

**Answer: B. Explanation: Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q67. Which statement uses Conservation agriculture without losing its vintage, basket or legal status?

A. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. B. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. C. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. D. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use.

**Answer: C. Explanation: Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q68. Which option avoids the standard UPSC close-option trap about Conservation agriculture?

A. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. B. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. C. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. D. Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system.

**Answer: D. Explanation: Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q69. Which statement correctly identifies Precision input tools?

A. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. B. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. C. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. D. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication.

**Answer: A. Explanation: Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q70. Which option preserves the accounting or regulatory boundary of Precision input tools?

A. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. B. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. C. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. D. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use.

**Answer: B. Explanation: Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q71. Which statement uses Precision input tools without losing its vintage, basket or legal status?

A. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. B. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. C. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. D. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning.

**Answer: C. Explanation: Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q72. Which option avoids the standard UPSC close-option trap about Precision input tools?

A. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. B. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. C. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. D. Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument.

**Answer: D. Explanation: Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q73. Which statement correctly identifies Sikkim organic transition?

A. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. B. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. C. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. D. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter.

**Answer: A. Explanation: Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q74. Which option preserves the accounting or regulatory boundary of Sikkim organic transition?

A. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. B. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. C. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. D. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use.

**Answer: B. Explanation: Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q75. Which statement uses Sikkim organic transition without losing its vintage, basket or legal status?

A. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. B. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. C. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. D. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning.

**Answer: C. Explanation: Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q76. Which option avoids the standard UPSC close-option trap about Sikkim organic transition?

A. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. B. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. C. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. D. Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication.

**Answer: D. Explanation: Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q77. Which statement correctly identifies Tenant inclusion?

A. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. B. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. C. Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. D. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged.

**Answer: A. Explanation: Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q78. Which option preserves the accounting or regulatory boundary of Tenant inclusion?

A. The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. B. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. C. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. D. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive.

**Answer: B. Explanation: Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q79. Which statement uses Tenant inclusion without losing its vintage, basket or legal status?

A. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. B. Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. C. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. D. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem.

**Answer: C. Explanation: Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

### Q80. Which option avoids the standard UPSC close-option trap about Tenant inclusion?

A. Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. B. An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. C. Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. D. Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter.

**Answer: D. Explanation: Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. The other options belong to different measures, vintages, baskets, instruments, institutions or legal perimeters.**

## PYQS AND ANSWER PRACTICE

### VERIFIED PYQ OWNERSHIP AUDIT

Audited ledgers route Mains demands on organic transition, integrated farming, watershed development, Jal Shakti Abhiyan, water storage, micro-irrigation, irrigation-system challenges and groundwater depletion. Objective demands cover conservation agriculture, DCCBs, KCC, biochar, zero tillage, fertigation, crop-protection chemicals and the provisionally keyed 2026 Rainfed Area Development question; no answer letter is inferred.

### OWNER PYQ LEDGER EXTRACTS

#### 9. PYQ application

- ANALYSIS: 2024 GS-III: Challenges of Indian irrigation and government measures for efficiency.
- ANALYSIS: 2025 GS-III: Factors for depleting groundwater and government steps.
- ANALYSIS: 2026 Prelims provisional key: Rainfed Area Development and Integrated Farming Systems.
- ANALYSIS: 2018 GS-III: Sikkim Organic State's ecological and economic benefits.
- ANALYSIS: 2019 GS-III: National Watershed Project's impact on water-stressed-area agriculture.
- ANALYSIS: 2020 GS-III: Jal Shakti Abhiyan's features for water conservation and security.
- ANALYSIS: **Irrigation answer route:** diagnose source depletion, groundwater dependence, uneven canal delivery, field-level inefficiency and distorted water-energy incentives; then deploy PMKSY, micro-irrigation, canal repair, aquifer management and crop alignment as coordinated responses.
- ANALYSIS: **Groundwater answer route:** connect free or subsidised power, paddy-sugarcane style water intensity, weak aquifer governance and limited crop diversification; then add Atal Jal, metering or budgeting logic, micro-irrigation and procurement reform.
- ANALYSIS: **Watershed/Jal Shakti answer route:** define the watershed unit, explain ridge-to-valley treatment and participatory institutions, then connect to Jal Shakti Abhiyan's five interventions and its 2021 "Catch the Rain" evolution as the campaign-mode complement to the standing watershed programme.
- ANALYSIS: **Sikkim organic-state answer route:** trace the phased 2003-2016 transition (resolution, subsidy withdrawal, sales ban, certification, declaration), then weigh ecological/economic benefits against replication limits for larger, plains-based states.
- ANALYSIS: **Unfamiliar-question route:** if asked on sustainable farming practices, connect conservation agriculture, zero tillage, fertigation, biochar, Integrated Farming Systems and climate-smart agriculture instead of writing a generic environment answer.

**Demand decoding:** The directive **answer** requires a direct position on "9. PYQ application", every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

#### Detailed examiner-grade model answer:

**Introduction and thesis:** The answer must resolve the Economy demand in "9. PYQ application".

#### Analytical body:

1. **Claim and named evidence:** ANALYSIS: 2024 GS-III: Challenges of Indian irrigation and government measures for efficiency. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** ANALYSIS: 2025 GS-III: Factors for depleting groundwater and government steps. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
3. **Claim and named evidence:** ANALYSIS: 2026 Prelims provisional key: Rainfed Area Development and Integrated Farming Systems. **Analysis:** Connect the identity/category and named Indian evidence -> instrument,

price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

4. **Claim and named evidence:** ANALYSIS: 2018 GS-III: Sikkim Organic State's ecological and economic benefits.

**Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

5. **Claim and named evidence:** ANALYSIS: 2019 GS-III: National Watershed Project's impact on

water-stressed-area agriculture. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

6. **Claim and named evidence:** ANALYSIS: 2020 GS-III: Jal Shakti Abhiyan's features for water conservation and

security. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion:** The answer must resolve the Economy demand in "9. PYQ application".

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For "9. PYQ application", replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

## 2026 PYQ Integration

**Status:** 2026 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: \_PYQ-ROUTING-PRELIMS-2026.md. **Answer-key rule:** The 2026 Prelims and CSAT Set-A keys held locally are **provisional**; no option or answer is recorded or inferred in this integration.

- **Year represented:** 2026
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 1

| Year | Paper        | Q  | PYQ demand (neutral rendering)                                            | Directive / format                                                                  | Source status                                                                                                                                | Owner requirement                                                         |
|------|--------------|----|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 2026 | Prelims GS-I | 28 | Rainfed Area Development objectives under sustainable agriculture mission | Objective question; provisional 2026 Set-A key present locally, answer not inferred | Provisional 2026 Set-A key present locally (Ans - 2026 - GS1 - Provisional); key is provisional - no answer letter recorded or inferred here | Cover the named fact/concept and its likely statement-level distinctions. |

### What this owner must now support

- Rainfed Area Development objectives under sustainable agriculture mission

This block integrates the 2026 examinable demand and paper metadata. It is kept separate from the 2018-2023 and 2024-2025 blocks and does not convert a provisionally-keyed, answer-free objective question into a solved answer.

### Recent PYQ Integration (2024-2025)

**Status:** 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: \_PYQ-ROUTING-MAINS-GS3-GS4-2024-2025.md.

- **Years represented:** 2024, 2025
- **Paper(s):** GS-III
- **Routed question demands:** 2

| Year | Paper  | Q  | PYQ demand (neutral rendering)                                     | Directive / format                    | Source status          | Owner requirement                                                                           |
|------|--------|----|--------------------------------------------------------------------|---------------------------------------|------------------------|---------------------------------------------------------------------------------------------|
| 2024 | GS-III | 13 | Challenges of the Indian irrigation system and government measures | State measures · 15 marks · 250 words | Routed to owning topic | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2025 | GS-III | 13 | Factors for depleting groundwater and government steps             | Examine · 15 marks · 250 words        | Routed to owning topic | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |

### What this owner must now support

- Challenges of the Indian irrigation system and government measures
- Factors for depleting groundwater and government steps

This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

### Historical PYQ Integration (2018-2023)

**Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: \_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md, \_PYQ-ROUTING-PRELIMS-2018-2023.md.

**Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2018, 2019, 2020, 2021, 2022
- **Paper(s):** GS-III, Prelims GS-I
- **Routed question demands:** 19

| Year | Paper        | Q  | PYQ demand (neutral rendering)                                        | Directive / format                                   | Source status                   | Owner requirement                                                                           |
|------|--------------|----|-----------------------------------------------------------------------|------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------|
| 2018 | GS-III       | 8  | Sikkim Organic State ecological and economic benefits                 | Discuss · 10 marks · 150 words                       | Routed to owning topic          | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2018 | Prelims GS-I | 59 | Conservation Agriculture practices minimum tillage and crop rotations | Objective question; official key unavailable locally | Routed; key unavailable locally | Cover the named fact/concept and its likely statement-level distinctions.                   |

| Year | Paper        | Q  | PYQ demand (neutral rendering)                                             | Directive / format                                   | Source status                                                                                    | Owner requirement                                                                           |
|------|--------------|----|----------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 2019 | GS-III       | 3  | Integrated Farming System role in sustaining agricultural production       | Discuss · 10 marks · 150 words                       | Cross-routed to sustainable-input and crop-livestock integration owners                          | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2019 | GS-III       | 4  | National Watershed Project impact on water-stressed area agriculture       | Elaborate · 10 marks · 150 words                     | Routed to owning topic                                                                           | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2019 | Prelims GS-I | 39 | Carbofuran phorate triazophos chemical use in agriculture                  | Objective question; official key unavailable locally | Routed; key unavailable locally                                                                  | Cover the named fact/concept and its likely statement-level distinctions.                   |
| 2020 | GS-III       | 8  | Jal Shakti Abhiyan features for water conservation and security            | What are · 10 marks · 150 words                      | Routed to owning topic                                                                           | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2020 | GS-III       | 14 | Measures to improve water storage and irrigation under depletion           | Suggest · 15 marks · 250 words                       | Routed to owning topic                                                                           | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2020 | Prelims GS-I | 59 | District Central Cooperative Banks agricultural credit delivery            | Objective question; official key unavailable locally | Routed; key unavailable locally                                                                  | Cover the named fact/concept and its likely statement-level distinctions.                   |
| 2020 | Prelims GS-I | 66 | Kisan Credit Card scheme short-term credit eligible purposes               | Objective question; official key unavailable locally | Routed; key unavailable locally                                                                  | Cover the named fact/concept and its likely statement-level distinctions.                   |
| 2020 | Prelims GS-I | 80 | Biochar uses in farming soil microorganisms and water retention            | Objective question; official key unavailable locally | Routed; key unavailable locally                                                                  | Cover the named fact/concept and its likely statement-level distinctions.                   |
| 2020 | Prelims GS-I | 83 | Zero tillage benefits for wheat paddy and carbon sequestration             | Objective question; official key unavailable locally | Routed; key unavailable locally                                                                  | Cover the named fact/concept and its likely statement-level distinctions.                   |
| 2020 | Prelims GS-I | 89 | Sugarcane cultivation bud chip settlings and tissue culture                | Objective question; official key unavailable locally | Routed; key unavailable locally                                                                  | Cover the named fact/concept and its likely statement-level distinctions.                   |
| 2020 | Prelims GS-I | 90 | Eco-friendly agriculture practices crop diversification tensiometer        | Objective question; official key unavailable locally | Routed; key unavailable locally                                                                  | Cover the named fact/concept and its likely statement-level distinctions.                   |
| 2020 | Prelims GS-I | 91 | Fertigation advantages alkalinity nutrient availability leaching reduction | Objective question; official key unavailable locally | Routed; key unavailable locally                                                                  | Cover the named fact/concept and its likely statement-level distinctions.                   |
| 2020 | Prelims GS-I | 94 | Chemical fertilizers ammonia source sulphur input and pricing              | Objective question; official key unavailable locally | Cross-routed to input fundamentals and fertiliser pricing/subsidy owner; key unavailable locally | Cover the named fact/concept and its likely statement-level distinctions.                   |
| 2021 | GS-III       | 4  | Micro-irrigation role and extent in solving India's water crisis           | How · 10 marks · 150 words                           | Routed to owning topic                                                                           | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |

| Year | Paper        | Q  | PYQ demand (neutral rendering)                                    | Directive / format                                   | Source status                                                               | Owner requirement                                                                           |
|------|--------------|----|-------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 2021 | Prelims GS-I | 51 | Permaculture farming versus conventional chemical farming         | Objective question; official key unavailable locally | Routed; key unavailable locally                                             | Cover the named fact/concept and its likely statement-level distinctions.                   |
| 2021 | Prelims GS-I | 59 | Climate-Smart Agriculture CCAFS CGIAR ICRISAT India               | Objective question; official key unavailable locally | Routed; key unavailable locally                                             | Cover the named fact/concept and its likely statement-level distinctions.                   |
| 2022 | GS-III       | 14 | Integrated Farming System benefits for small and marginal farmers | Explain · 15 marks · 250 words                       | Cross-routed to sustainable-input and smallholder animal-integration owners | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |

### What this owner must now support

- Sikkim Organic State ecological and economic benefits
- Conservation Agriculture practices minimum tillage and crop rotations
- Integrated Farming System role in sustaining agricultural production
- National Watershed Project impact on water-stressed area agriculture
- Carbofuran phorate triazophos chemical use in agriculture
- Jal Shakti Abhiyan features for water conservation and security
- Measures to improve water storage and irrigation under depletion
- District Central Cooperative Banks agricultural credit delivery
- Kisan Credit Card scheme short-term credit eligible purposes
- Biochar uses in farming soil microorganisms and water retention
- Zero tillage benefits for wheat paddy and carbon sequestration
- Sugarcane cultivation bud chip settlings and tissue culture
- Eco-friendly agriculture practices crop diversification tensiometer
- Fertigation advantages alkalinity nutrient availability leaching reduction
- Chemical fertilizers ammonia source sulphur input and pricing
- Micro-irrigation role and extent in solving India's water crisis
- Permaculture farming versus conventional chemical farming
- Climate-Smart Agriculture CCAFS CGIAR ICRISAT India
- Integrated Farming System benefits for small and marginal farmers

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

### 10. PYQ-based analytical application

- ANALYSIS: 2024 GS-III: Challenges of Indian irrigation and government measures for efficiency.
- ANALYSIS: 2026 Prelims provisional key: Rainfed Area Development and Integrated Farming Systems.
- ANALYSIS: **Irrigation answer engine:** source sustainability and aquifer governance; conveyance

and field efficiency; waterlogging/salinity; crop-water incentives and power pricing; then micro-irrigation, canal modernisation, watershed work, measurement, extension and locally credible user institutions.

### Historical PYQ Integration (2018-2023)

**Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: \_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md.

- **Years represented:** 2018, 2019, 2020, 2021, 2022

- Paper(s): GS-III

- Routed question demands: 7

| Year | Paper  | Q  | PYQ demand (neutral rendering)                                       | Directive / format               | Source status                                                               | Owner requirement                                                                           |
|------|--------|----|----------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 2018 | GS-III | 8  | Sikkim Organic State ecological and economic benefits                | Discuss · 10 marks · 150 words   | Routed to owning topic                                                      | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2019 | GS-III | 3  | Integrated Farming System role in sustaining agricultural production | Discuss · 10 marks · 150 words   | Cross-routed to sustainable-input and crop-livestock integration owners     | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2019 | GS-III | 4  | National Watershed Project impact on water-stressed area agriculture | Elaborate · 10 marks · 150 words | Routed to owning topic                                                      | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2020 | GS-III | 8  | Jal Shakti Abhiyan features for water conservation and security      | What are · 10 marks · 150 words  | Routed to owning topic                                                      | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2020 | GS-III | 14 | Measures to improve water storage and irrigation under depletion     | Suggest · 15 marks · 250 words   | Routed to owning topic                                                      | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2021 | GS-III | 4  | Micro-irrigation role and extent in solving India's water crisis     | How · 10 marks · 150 words       | Routed to owning topic                                                      | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |
| 2022 | GS-III | 14 | Integrated Farming System benefits for small and marginal farmers    | Explain · 15 marks · 250 words   | Cross-routed to sustainable-input and smallholder animal-integration owners | Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion. |

#### What this owner must now support

- Sikkim Organic State ecological and economic benefits
- Integrated Farming System role in sustaining agricultural production
- National Watershed Project impact on water-stressed area agriculture
- Jal Shakti Abhiyan features for water conservation and security
- Measures to improve water storage and irrigation under depletion
- Micro-irrigation role and extent in solving India's water crisis
- Integrated Farming System benefits for small and marginal farmers

The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

## PYQ DEMAND CARD 1 - 2018 GS-III

**Demand:** Discuss the ecological and economic benefits and limits of Sikkim's organic transition.

**Status:** Verified routed Mains demand; original model solution, not an official answer.

**Model solution: Sikkim organic transition:** Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Demand decoding:** The directive **answer** requires a direct position on "PYQ DEMAND CARD 1 - 2018 GS-III", every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Sikkim organic transition:** Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Analytical body:**

1. **Claim and named evidence:** Demand: Discuss the ecological and economic benefits and limits of Sikkim's organic transition. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand; original model solution, not an official answer. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Sikkim organic transition:** Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For "PYQ DEMAND CARD 1 - 2018 GS-III", replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

## PYQ DEMAND CARD 2 - 2019 GS-III

**Demand:** Elaborate the impact of watershed development on water-stressed agriculture.

**Status:** Verified routed Mains demand; original model solution, not an official answer.

**Model solution: Watershed method:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures.

**Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Rainfed Area Development:** Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Demand decoding:** The directive **answer** requires a direct position on “PYQ DEMAND CARD 2 - 2019 GS-III”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

### Detailed examiner-grade model answer:

**Introduction and thesis: Watershed method:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Rainfed Area Development:** Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

### Analytical body:

1. **Claim and named evidence:** Demand: Elaborate the impact of watershed development on water-stressed agriculture. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand; original model solution, not an official answer. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Watershed method:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures.

**Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Rainfed Area Development:** Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For “PYQ DEMAND CARD 2 - 2019 GS-III”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

### PYQ DEMAND CARD 3 - 2020 GS-III

**Demand:** Suggest measures for water storage and irrigation under groundwater depletion.

**Status:** Verified routed Mains demand; original model solution, not an official answer.

**Model solution: Potential and utilisation:** Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. **Access and efficiency:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. **Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Watershed method:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Atal Bhujal Yojana:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Demand decoding:** The directive **answer** requires a direct position on “PYQ DEMAND CARD 3 - 2020 GS-III”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Potential and utilisation:** Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. **Access and efficiency:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. **Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Watershed method:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Atal Bhujal Yojana:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Analytical body:**

1. **Claim and named evidence:** Demand: Suggest measures for water storage and irrigation under groundwater depletion. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand; original model solution, not an official answer. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Potential and utilisation:** Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. **Access and efficiency:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. **Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Watershed method:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Atal Bhujal Yojana:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For “PYQ DEMAND CARD 3 - 2020 GS-III”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

## PYQ DEMAND CARD 4 - 2021 GS-III

**Demand:** Assess the role and limits of micro-irrigation in addressing India's water crisis.

**Status:** Verified routed Mains demand; original model solution, not an official answer.

**Model solution: Access and efficiency:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. **Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Water-energy-crop nexus:** Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Demand decoding:** The directive **answer** requires a direct position on “PYQ DEMAND CARD 4 - 2021 GS-III”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Access and efficiency:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged.

**Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Water-energy-crop nexus:** Cheap or free pumping power lowers

private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

#### Analytical body:

1. **Claim and named evidence:** Demand: Assess the role and limits of micro-irrigation in addressing India's water crisis. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand; original model solution, not an official answer. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Access and efficiency:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged.

**Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Water-energy-crop nexus:** Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For "PYQ DEMAND CARD 4 - 2021 GS-III", replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

### PYQ DEMAND CARD 5 - 2024 GS-III

**Demand:** State the challenges of India's irrigation system and government measures.

**Status:** Verified routed Mains demand; original model solution, not an official answer.

**Model solution: Potential and utilisation:** Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. **PMKSY convergence:** The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. **Access and efficiency:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. **Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Watershed method:** Watershed development treats a geo-hydrological catchment from ridge to

valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Demand decoding:** The directive **answer** requires a direct position on “PYQ DEMAND CARD 5 - 2024 GS-III”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Potential and utilisation:** Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use.

**PMKSY convergence:** The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. **Access and efficiency:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. **Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Watershed method:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Analytical body:**

1. **Claim and named evidence:** Demand: State the challenges of India's irrigation system and government measures. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand; original model solution, not an official answer. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Potential and utilisation:** Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. **PMKSY convergence:** The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. **Access and efficiency:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. **Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Watershed method:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For “PYQ DEMAND CARD 5 - 2024 GS-III”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

### PYQ DEMAND CARD 6 - 2025 GS-III

**Demand:** Examine groundwater depletion and evaluate government responses.

**Status:** Verified routed Mains demand; original model solution, not an official answer.

**Model solution: Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Water-energy-crop nexus:** Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. **Atal Bhujal Yojana:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Rainfed Area Development:** Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Demand decoding:** The directive **answer** requires a direct position on “PYQ DEMAND CARD 6 - 2025 GS-III”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive.

**Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Water-energy-crop nexus:** Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. **Atal Bhujal Yojana:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Rainfed Area Development:** Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Analytical body:**

1. **Claim and named evidence:** Demand: Examine groundwater depletion and evaluate government responses. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** Status: Verified routed Mains demand; original model solution, not an official answer. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Micro-irrigation rebound:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive.

**Groundwater common pool:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Water-energy-crop nexus:** Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. **Atal Bhujal Yojana:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Rainfed Area Development:** Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For “PYQ DEMAND CARD 6 - 2025 GS-III”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

## ORIGINAL MAINS 1 - 10 MARKS

**Question:** Distinguish irrigation potential, utilisation and water-use efficiency. Answer in about 150 words.

**Model thesis: Claim:** Potential and utilisation. **Named evidence/example:** Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Claim -> named evidence -> analysis -> qualification:**

- Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use.

**Qualified conclusion: Claim:** Potential and utilisation. **Named evidence/example:** Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Demand decoding:** The directive **answer** requires a direct position on “Distinguish irrigation potential, utilisation and water-use efficiency. Answer in about 150...”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** Potential and utilisation. **Named evidence/example:** Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

### Analytical body:

1. **Claim and named evidence:** Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Claim:** Potential and utilisation. **Named evidence/example:** Created irrigation potential, utilised potential, reliable field delivery and water-use efficiency are different measures; project capacity does not prove equitable or productive use. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Executable exam-length answer / compression plan:** For 10 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For “Distinguish irrigation potential, utilisation and water-use efficiency. Answer in about 150...”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

### ORIGINAL MAINS 2 - 10 MARKS

**Question:** Separate irrigation access from precision application under PMKSY. Answer in about 150 words.

**Model thesis: Claim:** PMKSY convergence. **Named evidence/example:** The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:**

Access and efficiency. **Named evidence/example:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged.

**Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Claim -> named evidence -> analysis -> qualification:**

- The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning.
- Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged.

**Qualified conclusion: Claim:** PMKSY convergence. **Named evidence/example:** The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Access and efficiency. **Named evidence/example:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Demand decoding:** The directive **answer** requires a direct position on “Separate irrigation access from precision application under PMKSY. Answer in about 150 words.”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** PMKSY convergence. **Named evidence/example:** The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Access and efficiency. **Named evidence/example:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

#### **Analytical body:**

1. **Claim and named evidence:** The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Claim:** PMKSY convergence. **Named evidence/example:** The official PMKSY page describes convergence of field-level irrigation investment, assured irrigation expansion, on-farm efficiency, aquifer recharge and decentralised state planning. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and

geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Access and efficiency. **Named evidence/example:** Har Khet Ko Pani addresses irrigation access, while Per Drop More Crop addresses precise application and on-farm efficiency; access and efficiency must not be merged. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Executable exam-length answer / compression plan:** For 10 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For “Separate irrigation access from precision application under PMKSY. Answer in about 150 words.”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

### ORIGINAL MAINS 3 - 15 MARKS

**Question:** Explain the groundwater-energy-crop nexus and micro-irrigation rebound. Answer in about 250 words.

**Model thesis:** **Claim:** Micro-irrigation rebound. **Named evidence/example:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Groundwater common pool. **Named evidence/example:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Water-energy-crop nexus. **Named evidence/example:** Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Claim -> named evidence -> analysis -> qualification:**

- Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive.
- An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem.
- Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction.

**Qualified conclusion:** **Claim:** Micro-irrigation rebound. **Named evidence/example:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope,

implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Groundwater common pool. **Named evidence/example:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Water-energy-crop nexus. **Named evidence/example:** Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Demand decoding:** The directive **explain** requires a direct position on “Explain the groundwater-energy-crop nexus and micro-irrigation rebound. Answer in about 250...”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

#### Detailed examiner-grade model answer:

**Introduction and thesis:** **Claim:** Micro-irrigation rebound. **Named evidence/example:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Groundwater common pool. **Named evidence/example:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Water-energy-crop nexus. **Named evidence/example:** Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

#### Analytical body:

1. **Claim and named evidence:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
3. **Claim and named evidence:** Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. **Analysis:** Connect the

identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Claim:** Micro-irrigation rebound. **Named evidence/example:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Groundwater common pool. **Named evidence/example:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Water-energy-crop nexus. **Named evidence/example:** Cheap or free pumping power lowers private extraction cost and can reinforce water-intensive crops where procurement and other incentives point in the same direction. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For "Explain the groundwater-energy-crop nexus and micro-irrigation rebound. Answer in about 250...", replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

## ORIGINAL MAINS 4 - 15 MARKS

**Question:** How does watershed development differ from isolated water structures? Answer in about 250 words.

**Model thesis: Claim:** Watershed method. **Named evidence/example:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Jal Shakti campaign. **Named evidence/example:** The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period,

estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Atal Bhujal Yojana. **Named evidence/example:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Claim -> named evidence -> analysis -> qualification:**

- Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures.
- The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record.
- Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change.

**Qualified conclusion: Claim:** Watershed method. **Named evidence/example:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Jal Shakti campaign. **Named evidence/example:** The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Atal Bhujal Yojana. **Named evidence/example:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Demand decoding:** The directive **answer** requires a direct position on “How does watershed development differ from isolated water structures? Answer in about 250...”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

**Detailed examiner-grade model answer:**

**Introduction and thesis: Claim:** Watershed method. **Named evidence/example:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Jal Shakti campaign. **Named evidence/example:** The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Atal Bhujal Yojana. **Named evidence/example:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that

emphasises community participation, water budgeting and behavioural change. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

#### Analytical body:

1. **Claim and named evidence:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
2. **Claim and named evidence:** The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.
3. **Claim and named evidence:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Claim:** Watershed method. **Named evidence/example:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Jal Shakti campaign. **Named evidence/example:** The owner records Jal Shakti Abhiyan as a 2019 campaign that later evolved into Catch the Rain; dated coverage and asset claims require the relevant campaign record. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Atal Bhujal Yojana. **Named evidence/example:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Executable exam-length answer / compression plan:** For 15 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For “How does watershed development differ from isolated water structures? Answer in about 250...”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

## ORIGINAL MAINS 5 - 20 MARKS

**Question:** Evaluate farm credit and crop insurance through liquidity, eligibility, basis risk and claims. Answer in about 300 words.

**Model thesis:** **Claim:** KCC liquidity role. **Named evidence/example:** Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** NABARD and co-operatives. **Named evidence/example:** NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Insurance boundary. **Named evidence/example:** Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** PMFBY scope. **Named evidence/example:** PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Enrolment and claims. **Named evidence/example:** Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Basis risk. **Named evidence/example:** Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Tenant inclusion. **Named evidence/example:** Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Claim -> named evidence -> analysis -> qualification:**

- Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss.
- NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture.
- Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss.
- PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification.
- Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes.
- Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss.
- Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter.

**Qualified conclusion: Claim:** KCC liquidity role. **Named evidence/example:** Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** NABARD and co-operatives. **Named evidence/example:** NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Insurance boundary. **Named evidence/example:** Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** PMFBY scope. **Named evidence/example:** PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Enrolment and claims. **Named evidence/example:** Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Basis risk. **Named evidence/example:** Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Tenant inclusion. **Named evidence/example:** Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain

the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Demand decoding:** The directive **evaluate** requires a direct position on “Evaluate farm credit and crop insurance through liquidity, eligibility, basis risk and...”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects, trade-offs and a qualified conclusion.

#### **Detailed examiner-grade model answer:**

**Introduction and thesis:** **Claim:** KCC liquidity role. **Named evidence/example:** Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** NABARD and co-operatives. **Named evidence/example:** NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Insurance boundary. **Named evidence/example:** Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** PMFBY scope. **Named evidence/example:** PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Enrolment and claims. **Named evidence/example:** Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Basis risk. **Named evidence/example:** Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Tenant inclusion. **Named evidence/example:** Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

#### **Analytical body:**

1. **Claim and named evidence:** Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

2. **Claim and named evidence:** NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

3. **Claim and named evidence:** Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

4. **Claim and named evidence:** PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

5. **Claim and named evidence:** Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

6. **Claim and named evidence:** Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

7. **Claim and named evidence:** Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Claim:** KCC liquidity role. **Named evidence/example:** Kisan Credit Card is an institutional working-capital channel for crop and eligible allied needs; it addresses seasonal liquidity rather than compensating a realised insured loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** NABARD and co-operatives. **Named evidence/example:** NABARD is a rural and agricultural refinance and development institution, while District Central Cooperative Banks occupy a distinct tier in the cooperative credit architecture. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated

period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Insurance boundary. **Named evidence/example:** Crop insurance pools specified production or weather risk under scheme terms; it is not a universal guarantee of price, income, debt repayment or every individual loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** PMFBY scope. **Named evidence/example:** PMFBY operates through notified crops, areas, seasons and perils under the applicable scheme rules; current premium, enrolment and state-participation claims require the governing notification. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Enrolment and claims. **Named evidence/example:** Insurance enrolment or premium collection does not prove claim admissibility, assessment, settlement amount or timeliness; these are separate implementation outcomes. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Basis risk. **Named evidence/example:** Area-yield assessment reduces individual verification cost but can create basis risk when the reference-area loss differs from a particular farmer's field loss. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Tenant inclusion. **Named evidence/example:** Title-linked credit, insurance and subsidy administration can exclude tenants and sharecroppers even when they are the actual cultivators, so records and eligibility design matter. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Executable exam-length answer / compression plan:** For 20 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For "Evaluate farm credit and crop insurance through liquidity, eligibility, basis risk and...", replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.

## ORIGINAL MAINS 6 - 20 MARKS

**Question:** Design a sustainable agriculture strategy for rainfed and groundwater-stressed India. Answer in about 300 words.

**Model thesis:** **Claim:** Micro-irrigation rebound. **Named evidence/example:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Groundwater common pool. **Named evidence/example:** An individual pumper captures private benefit while cumulative extraction lowers a

shared aquifer, making groundwater a common-pool governance problem. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Watershed method. **Named evidence/example:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Atal Bhujal Yojana. **Named evidence/example:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Rainfed Area Development. **Named evidence/example:** Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Conservation agriculture. **Named evidence/example:** Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Precision input tools. **Named evidence/example:** Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Sikkim organic transition. **Named evidence/example:** Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Claim -> named evidence -> analysis -> qualification:**

- Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive.
- An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem.
- Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures.
- Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change.
- Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk.
- Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system.

- Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument.
- Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication.

**Qualified conclusion: Claim:** Micro-irrigation rebound. **Named evidence/example:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Groundwater common pool. **Named evidence/example:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Watershed method. **Named evidence/example:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Atal Bhujal Yojana. **Named evidence/example:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Rainfed Area Development. **Named evidence/example:** Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Conservation agriculture. **Named evidence/example:** Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Precision input tools. **Named evidence/example:** Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Sikkim organic transition. **Named evidence/example:** Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

**Demand decoding:** The directive **answer** requires a direct position on “Design a sustainable agriculture strategy for rainfed and groundwater-stressed India. Answer...”, every clause, exact formula/category, institution and policy status, a monetary-fiscal-real-external transmission chain with lags, named Indian evidence, distribution/federal effects,

trade-offs and a qualified conclusion.

#### Detailed examiner-grade model answer:

**Introduction and thesis:** **Claim:** Micro-irrigation rebound. **Named evidence/example:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Groundwater common pool. **Named evidence/example:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Watershed method. **Named evidence/example:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Atal Bhujal Yojana. **Named evidence/example:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Rainfed Area Development. **Named evidence/example:** Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Conservation agriculture. **Named evidence/example:** Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Precision input tools. **Named evidence/example:** Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Sikkim organic transition. **Named evidence/example:** Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source.

#### Analytical body:

1. **Claim and named evidence:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice remains water-intensive. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel ->

implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

2. **Claim and named evidence:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

3. **Claim and named evidence:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures.

**Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

4. **Claim and named evidence:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

5. **Claim and named evidence:** Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

6. **Claim and named evidence:** Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; zero tillage is one practice rather than the whole system. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect. **Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

7. **Claim and named evidence:** Fertigation applies nutrients through irrigation water, while a tensiometer measures soil-water tension for irrigation scheduling; neither is itself a crop-insurance instrument. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

8. **Claim and named evidence:** Sikkim's organic transition was phased through policy, input withdrawal, mission support and certification; its scale and hill ecology limit mechanical replication. **Analysis:** Connect the identity/category and named Indian evidence -> instrument, price, quantity, balance-sheet or incentive channel -> implementation and lag -> growth, employment, distribution, stability, sustainability and federal effect.

**Qualification:** State unit/denominator, stock/flow, nominal/real, authority and programme status, source-date-reference period, causal limit, counterfactual, trade-off or residual risk.

**Counter-position / limit:** An accounting identity, index movement, allocation, rate change, notification, platform count, WTO notification or chronological association cannot alone establish transmission, utilisation, distributional benefit or attributable outcome; test mechanism, lag, counterfactual, data vintage, implementation and external/federal constraints.

**Qualified conclusion: Claim:** Micro-irrigation rebound. **Named evidence/example:** Drip and sprinkler systems can reduce water applied per unit at plot level, but basin extraction may not fall if irrigated area expands or crop choice

remains water-intensive. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Groundwater common pool. **Named evidence/example:** An individual pumper captures private benefit while cumulative extraction lowers a shared aquifer, making groundwater a common-pool governance problem. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Watershed method. **Named evidence/example:** Watershed development treats a geo-hydrological catchment from ridge to valley through soil, moisture, recharge, vegetation and participatory maintenance rather than isolated structures. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Atal Bhujal Yojana. **Named evidence/example:** Atal Bhujal Yojana is a groundwater-management programme for water-stressed areas that emphasises community participation, water budgeting and behavioural change. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Rainfed Area Development. **Named evidence/example:** Rainfed Area Development under the National Mission for Sustainable Agriculture promotes integrated farming systems and diversification suited to rainfed risk. **Analysis:** This identifies the economic mechanism, the relevant institution or accounting boundary, and the effect on output, prices, distribution or financial stability. **Qualification:** The conclusion must retain the stated period, estimate vintage, base year, basket, instrument eligibility, crop and geography scope, implementation stage, stock-flow distinction or legal perimeter rather than generalise beyond the source. **Claim:** Conservation agriculture. **Named evidence/example:** Conservation agriculture combines minimum soil disturbance, residue cover and crop rotation; 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**Executable exam-length answer / compression plan:** For 20 marks, spend one-sixth of the time decoding the directive and drawing definition/identity -> instrument or shock -> transmission -> growth, distribution, stability, sustainability and federal effects; state a thesis; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, denominator, data vintage, legal/programme status, lag, causation and residual-risk checks.

**Why this earns marks:** The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves formula, stock-flow, institutional, status, data-vintage, distributional and causal distinctions.

**How to improve this answer:** For “Design a sustainable agriculture strategy for rainfed and groundwater-stressed India. Answer...”, replace the weakest catalogue point with one exact formula or classification, named institution/instrument, balance-sheet or incentive channel, lag, measurable outcome, distribution/federal effect and qualification.